

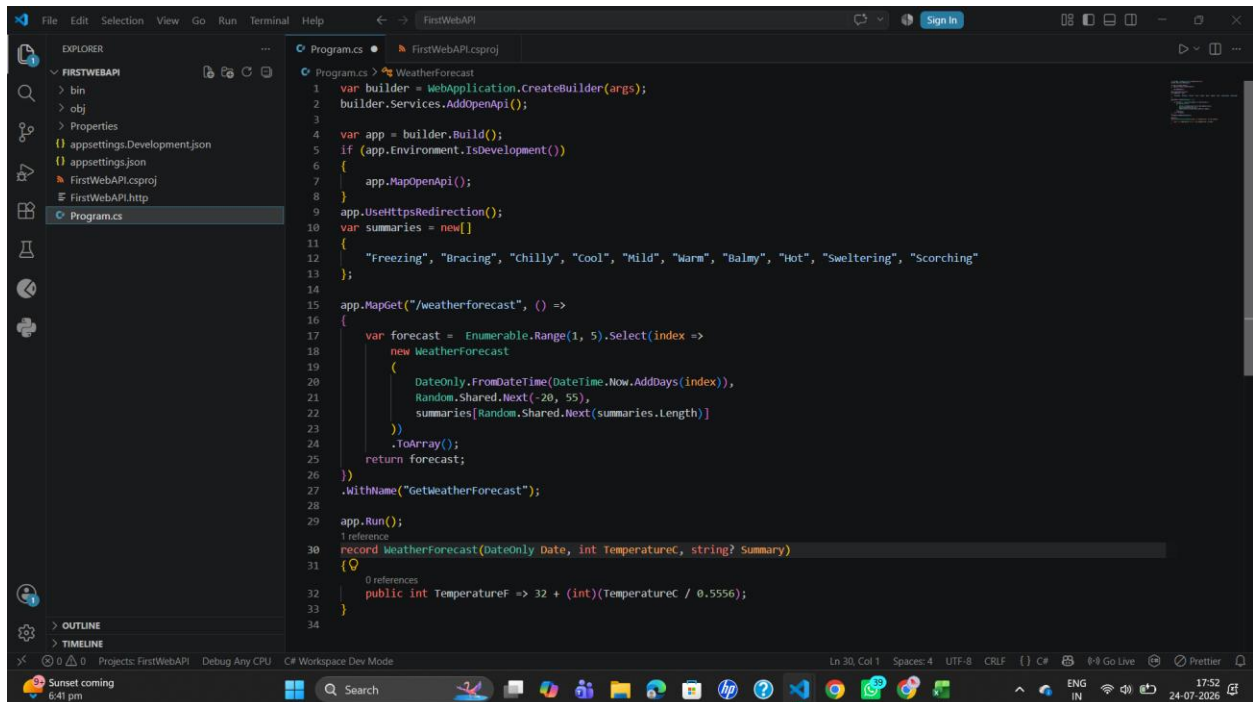
Lab 1: First ASP.NET Core Web API

Objective

To create a simple ASP.NET Core Web API, understand RESTful services, HTTP verbs, and execute the default GET endpoint using Swagger.

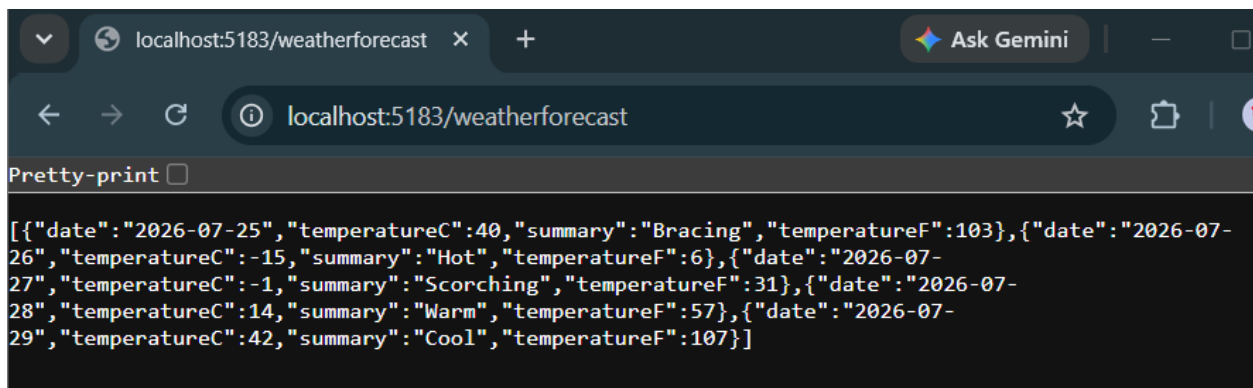
Code

Program.cs:



```
1 var builder = WebApplication.CreateBuilder(args);
2 builder.Services.AddOpenApi();
3
4 var app = builder.Build();
5 if (app.Environment.IsDevelopment())
6 {
7     app.MapOpenApi();
8 }
9 app.UseHttpsRedirection();
10 var summaries = new[]
11 {
12     "Freezing", "Bracing", "Chilly", "Cool", "Mild", "Warm", "Balmy", "Hot", "Sweltering", "Scorching"
13 };
14
15 app.MapGet("/weatherforecast", () =>
16 {
17     var forecast = Enumerable.Range(1, 5).Select(index =>
18         new WeatherForecast
19         (
20             DateOnly.FromDateTime(DateTime.Now.AddDays(index)),
21             Random.Shared.Next(-20, 55),
22             summaries[Random.Shared.Next(summaries.Length)]
23         ))
24         .ToArray();
25     return forecast;
26 });
27 app.MapGet("/weatherforecast", () =>
28 {
29     app.Run();
30     record WeatherForecast(DateOnly Date, int TemperatureC, string? Summary)
31     {
32         public int TemperatureF => 32 + (int)(TemperatureC / 0.5556);
33     }
34 });
35
36 app.Run();
```

Output



```
[{"date": "2026-07-25", "temperatureC": 40, "summary": "Bracing", "temperatureF": 103}, {"date": "2026-07-26", "temperatureC": -15, "summary": "Hot", "temperatureF": 6}, {"date": "2026-07-27", "temperatureC": -1, "summary": "Scorching", "temperatureF": 31}, {"date": "2026-07-28", "temperatureC": 14, "summary": "Warm", "temperatureF": 57}, {"date": "2026-07-29", "temperatureC": 42, "summary": "Cool", "temperatureF": 107}]
```